

Name of Student: _____

Scholar No.: _____

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Electronics and Communication Engineering

Mid-Term Examination Sep - 2025

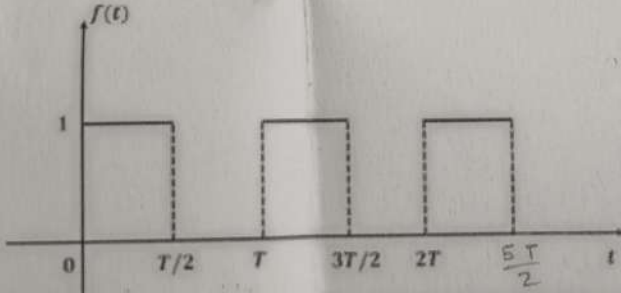
Subject Name/ Code:- SCS/CSE-3005

Time Allotted: 90 min

Max.Marks : 20

deepa Sharma

NOTE: 1. All questions are compulsory.
2. Mark answers number clearly.

✓ 1. A	Compare two methods of AM generation—square-law modulation and switching modulation in detail.	3
B. ✓	Define Laplace Transform and state its importance in analyzing communication signals.	2
2. A	A signal is represented by $x(t) = \begin{cases} 1 & t < 1 \\ 0 & t > 1 \end{cases}$ The fourier transform of the convolved signal $y(t)=x(2t)*x(t/2)$ is given by?	3
✓ B.	A DSB-SC signal is generated using the carrier $\cos(2\pi f_c t + \theta)$ and modulating signal $x(t)$. Derive the envelop of the DSB-SC signal.	2
✓	figure below is 	
B. ✓	Compare DSB-SC, SSB-SC, and VSB in terms of bandwidth, power efficiency, and receiver complexity.	2
4. A.	Determine the percentage power saved when carrier and one of the sidebands are suppressed in an AM wave modulated to a depth of 50%.	3
B.	Let $x(t) \rightarrow X(j\omega)$ be Fourier transform pair. Derive the Fourier transform of the signal $x(5t-3)$ in terms of $X(j\omega)$.	2